EXPLICIT HAMILTONIAN BALL PACKINGS: GENERAL CAPACITIES, HIGHER DIMENSIONS AND THE SIEGEL–YAO CRITERION

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Abstract. We realise symplectic ball packings by explicit Hamiltonian motions. For admissible data one definition gives a smooth compactly supported Hamiltonian $H[n,k,C,c,s]$ whose time-one map carries the closed balls $B(s_i,c_i)$, at arbitrary disjoint positions, into the open ball of capacity C with pairwise disjoint images; one algorithm outputs a Hamiltonian of the form (fixed cutoff)$\times$(rational polynomial) with the same property. In dimension $2n\geq 6$ the admissible data are those satisfying the volume and Gromov two-ball inequalities, so every packing allowed by Conjecture A of Siegel and Yao, proved in a 2026 preprint of OpenAI, is realised by an explicit flow. In dimension four they are the data with $\sum_i c_i^2 < C^2$ and $3c_i < C$; in particular $k\geq 9$ equal balls fill B^4 up to any $\varepsilon>0$, which proves Biran's theorem without J-holomorphic curves or Seiberg–Witten theory. The companion paper treats $k\geq 10$ equal balls in dimension four by a linear comparison function. The new ingredients here are a comparison datum for every concave profile, given by a finite rational iterate of a rearrangement contraction; a scalar comparison datum for the four-dimensional cubic model; an explicit area-preserving rearrangement of planar convex functions, lifted to toric fibres; a bordered handle whose Moser maps are flows of explicit fields in a finite atlas; a model over the diagonal curve $y_j^2=t^2-j$ with an explicit transfer to the ball; for one large ball, a closed-form shell return ($n\geq 4$) and a polynomial bubble ($n=3$); and a sound rational certificate for the finite output. The main results are verified in Lean 4.

1. Introduction

Points of $\mathbb{R}^{2n}=\mathbb{C}^n$ are $z=(z_1,\dots,z_n)$, $z_j=x_j+iy_j$, with $\omega_0=\sum_j dx_j\wedge dy_j$. The Hamiltonian vector field of $H(t,z)$ is defined by $\iota_{X_H}\omega_0=dH$, i.e. $X_H=(\partial_y H,-\partial_x H)$, and ϕ_H^t is its flow from time 0 to time t. We write $B(c)=\{\pi|z|^2\leq c\}$ for the closed ball of capacity c, $B^\circ(c)$ for its interior, and $B(s,c)=s+B(c)$.

Balls of capacities $c_1,\dots,c_k$ can be packed symplectically into $B^\circ(C)$ only if $\sum_i c_i^n < C^n$ (volume) and $c_i+c_j<C$ for $i\neq j$ (Gromov's two-ball theorem [Gro85]). In dimension four there are further obstructions [MP94, Hut11], and every packing problem reduces to inequalities indexed by exceptional classes [McD98, LL01, LL02]; Biran [Bir97] proved that $k\geq 9$ equal balls fill B^4 up to any $\varepsilon>0$. In dimension $2n\geq 6$ there are no further obstructions: this is Conjecture A of Siegel and Yao [SY25], proved in a 2026 preprint of OpenAI [Ope26]. These proofs do not write down a Hamiltonian moving prescribed balls; in [Ope26] the comparison function is a fixed point, and the planar maps realising the rearrangements and the relative Moser isotopies are obtained from existence statements. Paper I [Son26a] constructs explicit Hamiltonian full packings of B^4 by $k\geq 10$ equal balls. Here we realise, by the time-one map of one explicit Hamiltonian moving balls at prescribed positions into the target, every packing allowed by the two inequalities in dimension $2n\geq 6$, and in dimension four a range that contains all nearly full packings by $k\geq 9$ equal balls; and we give an algorithm whose output is a rational polynomial times a fixed cutoff.

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1.1. **Results.**

Definition 1.1 (Packing specification)**.** Let $n \geq 1$, $k \geq 0$, $C > 0$, $c = (c_1, \dots, c_k) \in (0, \infty)^k$ and $s = (s_1, \dots, s_k) \in (\mathbb{R}^{2n})^k$. A function $H \colon \mathbb{R} \times \mathbb{R}^{2n} \to \mathbb{R}$ satisfies $\mathrm{Pack}(n, k, C, c, s)$ if

(a) H is C^∞, and there is a compact set $\mathcal{S} \subset \mathbb{R}^{2n}$ with $H(t, z) = 0$ for all $t \in \mathbb{R}$ and all $z \notin \mathcal{S}$;
(b) for every i and every $z \in B(s_i, c_i)$ there is an integral curve $\gamma \colon \mathbb{R} \to \mathbb{R}^{2n}$ of X_H with $\gamma(0) = z$, and every integral curve $\gamma \colon [0, 1] \to \mathbb{R}^{2n}$ of X_H with $\gamma(0) = z$ satisfies $\gamma(1) \in B^\circ(C)$.

Under (a) the flow ϕ_H^t is global and symplectic, and (b) says $\phi_H^1(B(s_i, c_i)) \subset B^\circ(C)$.

Definition 1.2 (Admissible data)**.** Let $n \geq 2$, $k \geq 1$, $C > 0$ and $c \in (0, \infty)^k$. The data (n, k, C, c) are *admissible* if

$$\begin{aligned} n \geq 3 : &\qquad \textstyle\sum_i c_i^n < C^n \quad \text{and} \quad c_i + c_j < C \ \text{ for all } i \neq j; \\ n = 2 : &\qquad \textstyle\sum_i c_i^2 < C^2 \quad \text{and} \quad 3c_i < C \ \text{ for all } i. \end{aligned}$$

For $n \geq 3$ these are exactly the two necessary conditions. For $n = 2$ they define a sufficient range: by [MP94, McD98, LL01, LL02], closed balls of capacities c_i embed in $B^\circ(C)$ if $\sum_i c_i^2 < C^2$ and $\sum_i m_i c_i < dC$ for every exceptional class $(d; m_1, \dots, m_k)$ with $d \geq 1$; such a class has $m_i \geq 0$ and $\sum_i m_i = 3d - 1$, so $3c_i < C$ gives $\sum_i m_i c_i < dC$.

Theorem 1. *Let $n \geq 2$, $k \geq 1$, $C > 0$, $c_1, \dots, c_k > 0$ and $s_1, \dots, s_k \in \mathbb{R}^{2n}$ be such that the closed balls $B(s_i, c_i)$ are pairwise disjoint and (n, k, C, c) is admissible. Then:*

(i) the function $H = H[n, k, C, c, s]$ of Definition 3.6 is C^∞ on $\mathbb{R} \times \mathbb{R}^{2n}$, vanishes outside $\mathbb{R} \times \mathcal{S}$ for an explicit compact set $\mathcal{S}$ and for $t \notin (0, 1)$, and satisfies $\mathrm{Pack}(n, k, C, c, s)$*; in particular ϕ_H^1 maps each $B(s_i, c_i)$ into $B^\circ(C)$, and the images are pairwise disjoint;*
(ii) (L1) *the definition of $H[n, k, C, c, s]$ is the same for all such data and uses only the operations of level L1 in Definition 2.18;*
(iii) (L2, L3) *there is one oracle algorithm which, given n, k and names of the real inputs $C, c_1, \dots, c_k, s_1, \dots, s_k$ (a rational number is named by its constant sequence), halts on all such data and outputs an integer B and a polynomial $P \in \mathbb{Q}[t, z]$ such that $H_{\mathrm{fin}} = \beta_B \cdot P$ satisfies* $\mathrm{Pack}(n, k, C, c, s)$*. The output may depend on the names; for computable inputs, given by computable names, it is an ordinary algorithm.*

Here level L1 means that H is built by explicitly listed operations (arithmetic, elementary functions, integrals, unique solutions of explicit ODEs, inverses, case distinctions, finite searches; Definition 2.18). A *name* of a real number x is a sequence of rationals $q(p)$ with $|q(p) - x| \leq 2^{-p}$, and $\beta_B(z) = \prod_l \beta(\zeta_l / B)$ is a fixed smooth cutoff equal to one on the cube $[-B/2, B/2]^{2n}$ and vanishing outside $(-3B/4, 3B/4)^{2n}$, so that L2 is a finite description of a Hamiltonian and L3 a uniform algorithm producing it.

Parts (i) and (ii) are proved in §4, from the model theorems of Section 3 and the passage from embeddings to one Hamiltonian of Paper I; for $k \geq 10$ equal capacities in dimension four with $\frac{3}{4} < \rho\sqrt{k} < 1$, $\rho = c/C$, they are the linear branch of Paper I. Part (iii) is Theorem 5.3, proved in Section 5; its termination argument uses part (i).

Corollary 2 (The Siegel–Yao criterion by explicit flows)**.** *Let $n \geq 3$, $k \geq 1$, $C > 0$ and $c_i > 0$, and let $B(s_i, c_i) \subset \mathbb{R}^{2n}$, $1 \leq i \leq k$, be pairwise disjoint closed balls. There is a symplectic embedding of a neighbourhood of $\bigsqcup_i B(s_i, c_i)$ into $B^\circ(C)$ if and only if $\sum_i c_i^n < C^n$ and $c_i + c_j < C$ for all $i \neq j$. In that case ϕ_H^1 with $H = H[n, k, C, c, s]$ is such an embedding.*

Sufficiency is the main theorem of [Ope26]; Corollary 2 realises the embedding as the time-one map of an explicit Hamiltonian moving balls at prescribed positions.

Corollary 3 (Equal balls)**.** *Let $n \geq 2$, $\varepsilon \in (0,1)$, and $k \geq 2^n$ if $n \geq 3$, $k \geq 9$ if $n = 2$. Let $c > 0$, let $s_1, \dots, s_k$ be centres with $B(s_i, c)$ pairwise disjoint, and let $C > 0$ with $kc^n < C^n < kc^n/(1-\varepsilon)$. Then $H[n, k, C, (c, \dots, c), s]$ satisfies* $\mathrm{Pack}(n, k, C, (c, \dots, c), s)$*, and $kc^n/C^n > 1 - \varepsilon$.*

Thus, for $n \geq 3$, $k \geq 2^n$ equal balls fill any fraction below one of $B(C)$ by an explicit flow; for $2 \leq k < 2^n$ the two-ball inequality caps the fraction at $k/2^n$. For $n = 2$ the bound $k \geq 9$ is Biran's [Bir97]; here it comes from the condition $3c < C$, and for $k \leq 8$ full packings do not exist except for $k = 1, 4$ [MP94]. The case $n = 2$ gives a proof, without J-holomorphic curves or Seiberg–Witten theory, of Biran's theorem that B^4 is fully packed by k equal balls for every $k \geq 9$; the earlier proofs [Bir97, Bir99, BH13, BHO16] use them, directly or through the results they cite.

Corollary 4 (A cone of unequal balls in dimension four)**.** *Let $k \geq 1$ and $c_1, \dots, c_k > 0$ with $\max_i c_i \leq \frac{1}{3}\big(\sum_i c_i^2\big)^{1/2}$, let $\varepsilon \in (0,1)$ and $C_\varepsilon = \big(\sum_i c_i^2/(1-\varepsilon/2)\big)^{1/2}$. For all centres s_i with $B(s_i, c_i)$ pairwise disjoint, $H[2, k, C_\varepsilon, c, s]$ satisfies* $\mathrm{Pack}(2, k, C_\varepsilon, c, s)$*, and $\sum_i c_i^2/C_\varepsilon^2 = 1 - \varepsilon/2$; the images fill the fraction $1 - \varepsilon/2$ of the volume of $B(C_\varepsilon)$.*

Corollary 5 (The Epoch data)**.** *Let $k \geq 9$ be an integer, $\varepsilon \in (0,1)$, $s_i = (3i-3, 0, 0, 0) \in \mathbb{R}^4$, $R = (k/(1-\varepsilon/2))^{1/4}$ and $C = \pi R^2$. Then $H = H[2, k, C, (\pi, \dots, \pi), s]$ satisfies* $\mathrm{Pack}(2, k, C, (\pi, \dots, \pi), s)$*: its time-one map sends each closed unit ball $B(s_i, \pi)$ into $B^\circ(\pi R^2)$, and $k/R^4 = 1 - \varepsilon/2 > 1 - \varepsilon$.*

For $k \geq 10$ and $\varepsilon < \frac{7}{8}$ this is the corresponding corollary of Paper I, which uses the linear comparison datum; the other cases use the scalar comparison datum. The corollary is the input of the Epoch part of Theorem 5.3, which, together with Paper I, answers the mathematical part of the construction problem of Epoch AI [Epo26] (see Remark 5.5 for its numerical interface).

1.2. **The construction and what is new.** We follow the mechanism of [Ope26], and its sign convention $\iota_{X_Q}\omega_V = -dQ$ for fibre Hamiltonians, opposite to the convention for H. The model is a product $\Sigma_\ell \times \mathbb{C}^m$, $m = n-1$, of a compact piece of an affine curve with the fibre, with fibre moments $p_j = \pi|w_j|^2$. A ball $B(a) \subset \mathbb{C} \times \mathbb{C}^m$, sliced by the moments of its last m coordinates, is a family of discs of capacity $(a - |p|)_+$, $|p| = \sum_j p_j$. For capacities below $\frac{1}{2}$ the base is the diagonal curve $y_j^2 = t^2 - j$, and the base area above a fibre point is $H_{\mathrm{b}}(p) = A_\ell(1 - 2|p|)$ with $A_\ell \uparrow 2^m$; the limiting profile has integral $1/n! = \mathrm{vol}\, B^\circ(1)$. For a nearly full packing the deficit $P_{\mathrm{b}} = H_{\mathrm{b}} - \sum_i (a_i - |p|)_+$ has positive integral but is negative near $p = 0$, so the base discs cannot simply be placed side by side. One twists the bundle across an annulus by a Hamiltonian isotopy q of the fibre; with a function K of the moments this redistributes base area between moment levels, and the balls fit if

$$(K - P)\big(p(q(w))\big) \leq K\big(p(w)\big) - \gamma \qquad \text{for every fibre point } w,$$

for some $\gamma > 0$ and a minorant $P \leq P_{\mathrm{b}}$. In [Ope26], K is the fixed point of a contraction built from increasing rearrangements, the rearrangements are realised by isotopies, the inequality is turned into embeddings by Moser's method, and the result is transferred to the target. There are three cases: all capacities below $\frac{C}{2}$; one capacity at least $\frac{C}{2}$ with $n \geq 4$; and one capacity at least $\frac{C}{2}$ with $n = 3$.

The common isotopy from the prescribed balls and the Hamiltonian generating it are those of Paper I, recalled in Section 2; so are the cubic model, the four-dimensional handle and the return to the four-ball. The following ingredients are new here:

(1) a comparison datum for every concave profile on a rational polytope: a finite rational iterate of the rearrangement contraction of [Ope26, §2], with explicit gap, in place of its fixed point (Theorem 2.11);
(2) an explicit area-preserving rearrangement of a planar convex function, lifted to the fibres of a toric model with face collars (Theorem 2.10, proved in §S3);
(3) in dimension four, a scalar comparison datum for the cubic model that covers all admissible capacities, including $k = 9$ and unequal capacities, where a linear comparison function can fail (Theorem 3.1);
(4) a bordered handle whose Moser maps are time-one maps of explicit fields in a finite coordinate atlas, which proves the packing theorem in a model (Theorem 2.6) for $n \geq 3$ (§S6–S8);
(5) a model over the fixed diagonal curve $y_j^2 = t^2 - j$, an explicit instance of the complete-intersection model of [Ope26, §5.2], with a transfer to the ball by explicit flows (Theorem 3.2);
(6) for one large ball and $n \geq 4$, a mixed product model with a closed-form return to the shell about the reserved ball; for $n = 3$, the polynomial cubic model of [Ope26] with a quantitative area tail and a polynomial bubble that inserts the large ball (Theorem 3.3);
(7) an algorithm with finite output: a sound rational Euler certificate for a candidate polynomial, and termination proved from the explicit Hamiltonian (Theorem 5.3).

Definitions 1.1 and 1.2, Theorem 1(i), (iii) and Corollaries 2–5 are verified in Lean 4; the explicitness statement (ii) is checked by a metaprogram that is not itself verified (Section 6).

1.3. **Relation to previous work.** Packing stability in dimension four was extended by Biran [Bir99], Buse, Hind and Opshtein [BH11, BH13, BHO16] and Edtmair [Edt25]; Buse and Hind proved it in higher dimensions. Siegel and Yao [SY25] stated Conjecture A and proved a stabilised two-ball theorem and the invariance of four-dimensional ball packing problems under products with closed surfaces. Explicit constructions go back to Traynor [Tra95], including full packings of B^{2n} by d^n equal balls; symplectic folding and wrapping are in [Sch05b, Sch05a, Sch18], and Witt Nyström [WN24] embedded large neighbourhoods of zero sections of normal bundles. It uses the rearrangement contraction of [Ope26, §2], and the formal proof uses lemmas of the Lean library accompanying [Ope26]; the packing existence theorem of that preprint is not used.

1.4. **Organisation.** Section 2 states the comparison data and the toric mechanism, proved here, and the passage from embeddings to one Hamiltonian, recalled from Paper I. Section 3 describes the four-dimensional scalar branch, the three higher-dimensional models and the case distinction. Section 4 proves Theorem 1(i), (ii) and the corollaries, Section 5 proves the finite form and the algorithm, and Section 6 describes the formal verification. The supplement contains the complete proofs; numbers with the prefix S refer to it, and "Paper I, …" refers to [Son26a] and its supplement.

2. Comparison data, the toric mechanism and the Hamiltonian

This section states the comparison data for concave profiles and the packing theorem in a toric model, both proved in the supplement, and recalls from Paper I [Son26a] the toric balls, the twist and the passage from embeddings to one Hamiltonian. Statements recalled from Paper I carry the tag "Paper I" and have the same name there. Here $m = n - 1$; the comparison data serve the higher-dimensional models and, for $m = 1$, the scalar branch in dimension four.

Dependencies. Theorem 2.11 follows from Proposition 2.9 and Theorem 2.10; the proofs are in §S2 and §S3. Theorem 2.6 is Theorem S6.18, with the handle construction of §S6.

The termination part of Theorem 5.3 uses Theorem 1(i), proved in §4. The statements of the supplement of Paper I used in the proofs are reproduced in §SA.

Conventions. The dilation $z \mapsto \lambda z$ multiplies ω_0 and capacities by λ^2; conjugation by it normalises the target capacity to 1. In fibre coordinates w_j the moments are $p_j = \pi|w_j|^2$ and the angles $\theta_j = \arg w_j/2\pi \in \mathbb{R}/\mathbb{Z}$, so that the fibre form is $\omega_V = \sum_j dp_j \wedge d\theta_j$ away from the axes. Fibre Hamiltonians Q follow the sign of [Ope26], $\iota_{X_Q}\omega_V = -dQ$, opposite to the global one. In affine coordinates $\omega_{\mathrm{FS}} = dd^c \log(1+|z|^2)$ with $d^c = i(\bar\partial - \partial)/(4\pi)$; the map $z \mapsto z/\sqrt{\pi(1+|z|^2)}$ identifies this chart symplectically with $B^\circ(1)$. All cutoffs and clocks use the flat step S of Definition SA.1, all of whose derivatives vanish at 0 and 1.

2.1. **Toric balls and the twist.** A downward polytope is

$$\Delta = \{p \in \mathbb{R}^m_{\geq 0} : \langle b_\nu, p\rangle \leq \ell_\nu,\ 1 \leq \nu \leq N\}, \qquad b_\nu \in \mathbb{R}^m_{\geq 0}, \quad \ell_\nu > 0,$$

with toric domain $V_\Delta = \{w \in \mathbb{C}^m : p(w) \in \Delta\}$. The equalities $\langle b_\nu, p\rangle = \ell_\nu$ are its outer faces, and $\tau_\Delta\Delta$, $0 < \tau_\Delta < 1$, is a core that keeps the source caps away from them.

Definition 2.1 (Toric ball; Paper I)**.** In $\mathbb{C}^n = \mathbb{C} \times \mathbb{C}^m$, put $\varpi = \pi|z_{\mathrm{b}}|^2$. The ball of capacity a is $B(a) = \{\varpi + |p| \leq a\}$. At a fibre point with $|p| \leq a$ its base slice is a disc of capacity $a - |p|$, so its slice-height function is $(a - |p|)_+$.

Lemma 2.2 (The cap integral; Paper I)**.** *For $a \geq 0$, $\int_{\mathbb{R}^m_{\geq 0}} (a - |p|)_+ \, dp = a^{m+1}/(m+1)!$.*

Proof. The simplex $\{|p| \leq t\}$ has measure $t^m/m!$; integrate its distribution function against $(a-t)_+$. □

Each model has a compact affine core Σ_ℓ and a horizontal primitive $\Gamma_{\mathrm{phys}}(p)$; the physical form is $\Omega = \omega_V + d\Gamma_{\mathrm{phys}}$, with base curvature $\kappa(p) = d_\Sigma \Gamma_{\mathrm{phys}}(p)$. The base profile and budget are

$$H_{\mathrm{b}}(p) = \int_{\Sigma_\ell} \kappa(p), \qquad \mathcal{B}_\Delta = \int_\Delta H_{\mathrm{b}}(p)\, dp. \tag{1}$$

For auxiliary capacities a_i larger than the given ones, with $\{|p| \leq a_i\} \subset \tau_\Delta\Delta$, put $P_{\mathrm{b}}(p) = H_{\mathrm{b}}(p) - \sum_i (a_i - |p|)_+$. By the cap integral,

$$\int_\Delta P_{\mathrm{b}} = \mathcal{B}_\Delta - \sum_i \frac{a_i^n}{n!}, \qquad \frac{1 - \sum_i \rho_i^n}{n!} = \Big(\frac{1}{n!} - \mathcal{B}_\Delta\Big) + \int_\Delta P_{\mathrm{b}} + \sum_i \frac{a_i^n - \rho_i^n}{n!}. \tag{2}$$

For small balls the model parameters make the first and last terms on the right smaller than the left side, so $\int_\Delta P_{\mathrm{b}} > 0$; for one large ball of reserved capacity a the limiting budget for the remaining balls is $(1 - a^n)/n!$. A positive integral does not give the pointwise inequality $P_{\mathrm{b}} \geq 0$ needed to stack the base discs; the twist uses only the integral.

The twist. The handle data (Definition S6.1) include an annulus with coordinates $(s, v) \in [0,1] \times \mathbb{R}/\mathbb{Z}$, a closed one-form $\beta = b(v)\, dv$ with $b \geq 0$, $\int_0^1 b = 1$, and a positive reference area form Λ. A first Moser path replaces the physical curvature on the annulus by a background $e_\Lambda \Lambda$ plus the base mass $H^*(p) \geq H_{\mathrm{b}}(p) - \gamma_*/16$ carried by β (§S6.1.1), where γ_* is a reserve set aside from the gap below. Let q_t be a Hamiltonian isotopy of the fibre, $q = q_1$; the suspension $\mathcal{D}_\lambda(s, v, w) = (s, v, q_{\lambda f(s)} w)$, $f = S$, inserts it across the annulus. For a function K of the moments put $J = K - H^*$; adding $d(J(p)\beta)$ changes only toric mixed terms away from the annulus, and after the twist the base area available across the annulus at w is

$$K\big(p(q^{-1}w)\big) - J\big(p(w)\big) = H^*\big(p(w)\big) + K\big(p(q^{-1}w)\big) - K\big(p(w)\big). \tag{3}$$

Since q preserves the fibre volume, this redistributes area between moment levels. A comparison supplies, for a minorant $P \le P_{\mathrm{b}}$,

$$(K-P)\big(p(qw)\big) \le K\big(p(w)\big) - \gamma \qquad (w \in V_\Delta), \qquad 0 < \gamma_* \le \gamma. \tag{4}$$

Applied at $q^{-1}w$ it shows that the available area exceeds $\sum_i (a_i - |p(w)|)_+$ by at least $\gamma - \gamma_*/16 > 0$. This is divided among disjoint *layers* $I_1, \dots, I_k, I_*$ of $s \in (0,1)$: across I_i the β-coefficient rises by a smooth cap $h_i(p)$ below $(a_i - |p|)_+$, and a ball of capacity slightly less than a_i is placed there in product form; across I_* it rises by the positive remainder. The general comparator is built from a concave piecewise affine minorant P with positive mean, with

$$0 < d < \min\big(\bar{P}, \beta_P\big), \qquad \bar{P} = |\Delta|^{-1} \int_\Delta P, \tag{5}$$

where $P \ge \beta_P > 0$ off the core $\tau_\Delta \Delta$ (§2.3).

2.2. **Model, comparison datum and transfer.**

Definition 2.3 (Model input)**.** A higher-dimensional model input consists of a compact affine core $\Sigma_\ell \subset M$ of an affine curve M, the product $E = M \times \mathbb{C}^m$, a downward polytope Δ, and the physical primitive $\Gamma_{\mathrm{phys}}(p)$ given in the corresponding model section, together with the handle data of Definition S6.1. The physical form is

$$\Omega = \omega_V + d\Gamma_{\mathrm{phys}}, \qquad V_\Delta(E) = \Sigma_\ell \times V_\Delta.$$

The physical curvature $\kappa(p) = d_\Sigma \Gamma_{\mathrm{phys}}(p)$ is positive on the model region and extends smoothly in the moment variables across the coordinate hyperplanes.

Definition 2.4 (Comparison datum)**.** A comparison datum for (Δ, P) with gap $\gamma > 0$ and reserve $\gamma_* \in (0, \gamma]$ consists of a function K of the moments, smooth on a neighbourhood of Δ in $\mathbb{R}^m$, and a smooth Hamiltonian isotopy $(q_t)_{t\in[0,1]}$ of $\mathbb{C}^m$ with $q_0 = \mathrm{id}$, generated by Q_t with $\iota_{X_{Q_t}} \omega_V = -dQ_t$, such that:

1. $q_t(V_\Delta) = V_\Delta$ for every t, and there is $\varsigma > 0$ such that for every face ν and every t, Q_t Poisson-commutes with the face moment $\mu_\nu = \langle b_\nu, p\rangle$ on the collar $\{\mu_\nu > \ell_\nu - \varsigma\}$;
2. $q = q_1$ satisfies the comparison inequality (4).

For the comparator of Theorem 2.11, with gap $\gamma = \frac{11}{16} d$, the models take the reserve $\gamma_* = \frac{11}{64} d$.

Definition 2.5 (Transfer)**.** A transfer to an open target $U \subset \mathbb{C}^n$ assigns to a compact subset $\mathcal{K}$ of the model domain a smooth symplectic embedding $\mathcal{T}_\mathcal{K}$ on a neighbourhood of $\mathcal{K}$, and an explicit compact set $\mathcal{C}_\mathcal{K} \subset U$ containing the image of $\mathcal{K}$. The cubic, quadric and six-dimensional polynomial transfers are fixed maps; in the shell transfer for $n \ge 4$ the slit-shell parameters depend on $\mathcal{K}$.

Theorem 2.6 (Packing in a model)**.** *Let $n \ge 3$ and let $(\Sigma_\ell, \Delta, \Gamma_{\mathrm{phys}})$ be a model input (Definition 2.3) whose handle data satisfy the collar bounds (S43). Let $(K, (q_t))$ be a comparison datum for (Δ, P) with gap γ and reserve γ_* (Definition 2.4). Let $0 < \rho_i < a_i$ $(1 \le i \le k)$ with $\{|p| \le a_i\} \subset \tau_\Delta \Delta$, and suppose*

$$P(p) \le H_{\mathrm{b}}(p) - \sum_i (a_i - |p|)_+ \qquad (p \in \Delta).$$

Then the maps ψ_i of Construction S8.4 are smooth symplectic embeddings of neighbourhoods of the closed balls $B(\rho_i)$ into (E, Ω), and the sets $\psi_i(B(\rho_i))$ are pairwise disjoint and contained in $\operatorname{int} V_\Delta(E)$. *For $n = 2$ the same conclusion holds for the cubic model of Paper I with its handle data and equivariant maps, when the comparison datum is replaced by valid cubic data (Paper I, Definition (Cubic data)): the full-disc comparison for the profile itself and the collar condition $P_{\mathrm{supp}} < h$.*

The maps are $\psi_i = \Phi^{-1} \circ \mathcal{D}^{-1} \circ \mathcal{E}_i$: a layer map $\mathcal{E}_i$ placing the ball in the layer I_i, the inverse twist, and the inverse of three Moser maps (§S6.6). The proof for the bordered base is Theorem S6.18, with Propositions S7.7 and S7.9. The model sections verify the density floor, the collar bounds and the profile minorant before applying the theorem.

Proposition 2.7 (Assembly)**.** *Let ψ_i be the model embeddings of Theorem 2.6 and let $\mathcal{T}$ be its transfer. Then, with $\mathcal{K} = \bigcup_i \psi_i(B(\rho_i))$, the maps $f_i = \mathcal{T}_{\mathcal{K}} \circ \psi_i$ are smooth symplectic embeddings on neighbourhoods of $B(\rho_i)$, have pairwise disjoint images in U, and have images contained in $\mathcal{C}_{\mathcal{K}}$.*

Proof. $\mathcal{K}$ is compact and the transfer is one injective symplectic map on a neighbourhood of it, so composition preserves the pullback identity and disjointness; the inverse images of that neighbourhood under the ψ_i are open neighbourhoods of the source balls. □

input	base and physical primitive	base profile	target
$n = 2$ (Paper I)	disc bundle over the cubic, on its periodic cover	$3(1 - 3p)$	$B^\circ(1)$
$n \geq 3$, all $c_i < C/2$	$y_j^2 = t^2 - j$, $m = n - 1$; $(1 - 2\lvert p\rvert)\lambda$	$A_\ell(1 - 2\lvert p\rvert)$; limit $2^m(1 - 2\lvert p\rvert)$	$B^\circ(1)$
$n \geq 4$, one large ball	the curve $y_j^2 = t^2 - j$, $j \leq n - 2$, fibre $\mathbb{C}^{n-1}$; $(1 - u - 2w)\lambda$	$A_\ell(1 - u - 2w)$; limit $2^{n-2}(1 - u - 2w)$	a shell about the reserved ball
$n = 3$, one large ball	an explicit polynomial primitive on a compact affine core	H_ℓ, limit $9(b - \lvert p\rvert) - 2(s - \lvert p\rvert)_+$	common polynomial form, then $B^\circ(1)$

TABLE 1. The models. A_ℓ is the reference area of the compact affine core; each higher-dimensional total space is a product over that core. For one large ball, $u = p_1$ and $w = p_2 + \cdots + p_{n-1}$ ($n \geq 4$); for $n = 3$, H_ℓ is the base profile of the truncated polynomial model, with limit H_0, and $s < b$ are explicit parameters of that model (§S11.2).

2.3. **Comparison data.** Let Δ be a compact full-dimensional downward polytope and P a continuous concave function on Δ with positive mean. Norms are uniform norms on Δ. For a coordinate j write $p = (y, x)$ with $x = p_j$; then $\Delta = \{(y, x) : y \in \Delta_j,\ 0 \leq x \leq L(y)\}$ with $L(y) = \min_{\nu:\, b_{\nu j} > 0}(\ell_\nu - \sum_{i \neq j} b_{\nu i} y_i)/b_{\nu j}$ concave piecewise affine on the projection Δ_j. The polytope is *rational* if all b_ν, ℓ_ν are rational; a function on Δ is *rational convex polyhedral* (resp. *rational concave piecewise affine*) if it is the maximum (resp. minimum) of finitely many affine functions with rational coefficients.

Definition 2.8 (Coordinate rearrangement)**.** For a continuous convex function F on Δ and a coordinate j, put

$$(R_j F)(y, x) = \min_{0 \leq a,\ a + x \leq L(y)} \max\big(F(y, a),\, F(y, a + x)\big), \qquad (y, x) \in \Delta,$$

and $\mathcal{R} = R_m \circ \cdots \circ R_1$.

Thus $x \mapsto (R_j F)(y, x)$ is the increasing rearrangement of $F(y, \cdot)$ on $[0, L(y)]$; its properties (monotonicity, nonexpansiveness, and preservation of rational convex polyhedral functions by Fourier–Motzkin elimination) are Lemma S2.2.

Proposition 2.9 (Finite comparator)**.** *Let Δ be rational, $\tau_\Delta \in (0,1)$ rational, and P a rational concave piecewise affine function on Δ with mean $\mu = \mathrm{vol}(\Delta)^{-1}\int_\Delta P > 0$ and $P \geq \beta_P$ on $\Delta \setminus \tau_\Delta\Delta$ for some $\beta_P > 0$. Let d be rational with $0 < d < \min(\mu, \beta_P)$ and put*

$$\eta_\Delta = (1-\tau_\Delta)(\mu - d), \qquad M_P = \max\bigl(1,\ d - \min_\Delta P + 1\bigr), \qquad \lambda = 1 - \frac{\eta_\Delta}{2(M_P + \eta_\Delta)}.$$

Define $K_0 = 0$, $K_{r+1} = \mathcal{R}(\lambda K_r - P)$ and

$$N = \left\lceil \frac{16\|P\|}{d(1-\lambda)(\lambda^{-1}-1)} \right\rceil + 1.$$

Then every K_r is rational convex polyhedral, $\lambda^N\|P\|/(1-\lambda) < d/16$, and

$$\mathcal{R}(K_N - P) \leq K_N - \tfrac{7}{8}d \qquad \text{on } \Delta.$$

All quantities are computed by finite rational arithmetic and Fourier–Motzkin elimination. The construction, and the smoothing and realisation below, apply verbatim over real coefficients ("convex polyhedral" in place of "rational convex polyhedral"); this version is used for real capacities.

Theorem 2.10 (Realisation of one coordinate rearrangement)**.** *Let Δ be rational, F rational convex polyhedral on Δ, $j \in \{1,\dots,m\}$ and $e > 0$ rational. Let $Q_t^{(j)}$ be the Hamiltonian defined from (Δ, F, j, e) in Definition S3.10, and let $(q_t^{(j)})_{t\in[0,1]}$ be its flow ($\iota_X\omega_V = -dQ_t^{(j)}$). Then $Q^{(j)}$ is smooth with compact support in w, $(q_t^{(j)})$ is a smooth Hamiltonian isotopy of $\mathbb{C}^m$, and:*

- *(a) $q_0^{(j)} = \mathrm{id}$; every $q_t^{(j)}$ maps V_Δ onto itself and preserves the moments p_i, $i \neq j$.*
- *(b) There is a rational $\varsigma_j > 0$ such that for every ν with $b_{\nu j} > 0$ and every t, $Q_t^{(j)}$ vanishes on $\{\langle b_\nu, p\rangle > \ell_\nu - \varsigma_j\}$; for every ν with $b_{\nu j} = 0$, $Q_t^{(j)}$ Poisson-commutes with $\langle b_\nu, p\rangle$.*
- *(c) $F\bigl(p(q_1^{(j)}(w))\bigr) \leq (R_jF)(p(w)) + e$ for every $w \in V_\Delta$.*

The proof is Theorem S3.13. The construction is planar: fix y and rescale an inner slice $[0, \ell_{\mathrm{in}}(y)]$, $\ell_{\mathrm{in}}(y) < L(y)$, to $[0,1]$; then $f(v) = F(y, \ell_{\mathrm{in}}(y)v)$ is convex with increasing rearrangement $f^\sharp$, and one needs an area-preserving map q_{in} of the disc with $f(\pi|q_{\mathrm{in}}(z)|^2) \leq f^\sharp(\pi|z|^2) + \xi$ on a core disc. The sublevel sets of a convex potential approximating $f(v)$ (log-sum-exp smoothing plus a barrier), proper in the angle and exactly quadratic near its minimum, give such a map through Morse charts (Theorem S3.7). A compactly supported Hamiltonian isotopy of the disc ends at this map on a core (Theorem SA.13); its constants are uniform in y (Theorem S3.9), and the planar generator is lifted as

$$Q_t^{(j)}(w) = \chi(y)S'(t)\ell_{\mathrm{in}}(y)\, Q^{\mathrm{pl}}\bigl(y, \chi(y)S(t), w_j/\sqrt{\ell_{\mathrm{in}}(y)}\bigr),$$

with a cutoff $\chi(y)$ vanishing near degenerate slices.

Theorem 2.11 (Comparison datum for concave profiles)**.** *Assume the hypotheses of Proposition 2.9, and let d, N and K_N be as there. Let $K_N = \max_{\iota\in I}\phi_\iota$ be the representation by rational affine functions produced by the iteration, each step R_j being the Fourier–Motzkin elimination of Lemma S2.2(v), set $c_K = \lceil 32\,\#I/d\rceil + 1$ (so that $\log(\#I)/c_K < d/32$ by Lemma S2.4), and put*

$$K = \frac{1}{c_K}\log\sum_{\iota\in I} e^{c_K\phi_\iota}.$$

Let $F_0 = K_N - P$ and $F_j = R_jF_{j-1}$ ($1 \leq j \leq m$); let $q_t^{(j)}$ be the isotopy of Theorem 2.10 for F_{j-1}, the coordinate j and $e_j = d/(8(m+1))$. Define q_t by running $q^{(m)}, q^{(m-1)}, \dots, q^{(1)}$ in turn

with flat clocks: for $t \in [\frac{r-1}{m}, \frac{r}{m}]$, $r = 1, \dots, m$, *and* $j = m + 1 - r$,

$$q_t = q^{(j)}_{S(mt-r+1)} \circ q_1^{(j+1)} \circ \cdots \circ q_1^{(m)}.$$

Then $(K, (q_t))$ *is a comparison datum for* (Δ, P) *with gap* $\gamma = \frac{11}{16} d$.

The proof chains the realisation errors e_j along $u_m = w$, $u_{j-1} = q_1^{(j)}(u_j)$, which gives $(K_N - P)(p(q_1(w))) \le K_N(p(w)) - \frac{3}{4}d$, and uses $K_N \le K < K_N + d/32$.

2.4. **From embeddings to one Hamiltonian.** The models give explicit symplectic embeddings of slightly enlarged balls centred at the origin into $B^\circ(1)$, with pairwise disjoint images. Paper I joins the translations $u \mapsto s_i + u$ to such embeddings by one explicit family of symplectic embeddings with pairwise disjoint images at every time, and writes down a Hamiltonian generating it on the moving balls. Both steps hold in every dimension $n \ge 1$.

Definition 2.12 (Normalisation; Paper I)**.** Let $C > 0$ and $c \in (0, \infty)^k$. The *normalised capacities* are $\rho_i = c_i / C$. For a map $f^\natural$ defined on a subset of $\mathbb{R}^{2n}$, its *physical rescaling* is $f(u) = \sqrt{C}\, f^\natural(u/\sqrt{C})$.

Admissibility is homogeneous, so (n, k, C, c) is admissible if and only if $(n, k, 1, \rho)$ is; the physical rescaling is symplectic, injective or of disjoint images exactly when $f^\natural$ is.

Definition 2.13 (Embedding datum; Paper I)**.** Let $n \ge 1$, $k \ge 1$, $C > 0$ and $c \in (0, \infty)^k$. An *embedding datum* for (n, k, C, c) consists of numbers $c_i', \bar{c}_i$ with $c_i < c_i' < \bar{c}_i$ and maps $f_i \colon B^\circ(\bar{c}_i) \to \mathbb{R}^{2n}$ such that

(i) f_i is C^∞ and $Df_i(u) \in \mathrm{Sp}(2n)$ for all $u \in B^\circ(\bar{c}_i)$;
(ii) f_i is injective on $B(c_i')$;
(iii) the sets $f_i(B(c_i'))$, $1 \le i \le k$, are pairwise disjoint and contained in $B^\circ(C)$.

Given an embedding datum and centres s_i with $B(s_i, c_i)$ pairwise disjoint, Paper I fixes radii $r_i = \sqrt{c_i/\pi} < r_{1,i} < r_{2,i} < r_{3,i} < r_{4,i} < r_i' = \sqrt{c_i'/\pi}$ (Paper I, Lemma (Radii)) and defines an isotopy $F_{i,t}$ by explicit stages: a collision-free route of translations, a path in $\mathrm{Sp}(2n)$ from I to $Df_i(0)$, a shape stage, and the Alexander tail $u \mapsto v^{-1} f_i(vu)$ (Paper I, Construction (The common isotopy)). Here $\mathbb{B}(y, r)$ and $\mathbb{B}^\circ(y, r)$ are the closed and open Euclidean balls, and $\mathbb{B}(r) = \mathbb{B}(0, r)$.

Proposition 2.14 (The common symplectic isotopy; Paper I)**.** *The isotopy* $F_{i,t}$ *has the following properties.*

(a) $(t, u) \mapsto F_{i,t}(u)$ *is* C^∞ *on* $\mathbb{R} \times \mathbb{B}^\circ(r_{4,i})$; $F_{i,0}(u) = s_i + u$, $F_{i,1} = f_i$; $F_{i,t} = F_{i,0}$ *for* $t \le 0$ *and* $F_{i,t} = F_{i,1}$ *for* $t \ge 1$;
(b) $D_u F_{i,t}(u) \in \mathrm{Sp}(2n)$ *for all* t *and* $u \in \mathbb{B}^\circ(r_{4,i})$;
(c) each $F_{i,t}$ *is injective on* $\mathbb{B}(r_{3,i})$, *and for every* t *the sets* $F_{i,t}(\mathbb{B}(r_{3,i}))$, $1 \le i \le k$, *are pairwise disjoint;*
(d) $F_{i,t}(\mathbb{B}(r_{3,i})) \subset \mathbb{B}(R_F)$ *for all* i, t, *for an explicit radius* R_F.

Definition 2.15 (The Hamiltonian of an embedding datum; Paper I)**.** For $u \in \mathbb{B}^\circ(r_{4,i})$ let $\mathfrak{a}_{i,t}(u)[w] = \omega_0\big(\dot{F}_{i,t}(u), D_u F_{i,t}(u) w\big)$ and $\mathfrak{h}_{i,t}(u) = \int_0^1 \mathfrak{a}_{i,t}(\vartheta u)[u]\, d\vartheta$. Let $\chi_i = \chi_{r_{1,i}, r_{2,i}}$ be the radial cutoff of Definition SA.3. Put $H_i(t, y) = \chi_i(u)\, \mathfrak{h}_{i,t}(u)$ if u is the unique point of $\mathbb{B}(r_{3,i})$ with $F_{i,t}(u) = y$, and $H_i(t, y) = 0$ if there is no such unique point; and $H_{\mathfrak{E}} = \sum_{i=1}^k H_i$, where $\mathfrak{E}$ denotes the embedding datum together with the centres.

Theorem 2.16 (The Hamiltonian realises the isotopy; Paper I)**.** *Let $H = H_{\mathfrak{E}}$. Then H is C^∞, vanishes for $t \notin (0,1)$ and outside $\mathbb{R} \times \mathbb{B}(R_F)$; the flow ϕ_H^t is defined for all $t \in \mathbb{R}$ and consists of symplectic diffeomorphisms of $\mathbb{R}^{2n}$, and*

$$\phi_H^t(s_i + u) = F_{i,t}(u) \qquad (u \in \mathbb{B}(r_i),\ t \in \mathbb{R},\ 1 \le i \le k).$$

In particular $\phi_H^1(B(s_i, c_i)) = f_i(B(c_i))$, and these sets are pairwise disjoint and contained in $B^\circ(C)$.

This combines Paper I, Lemma (Smoothness and support) with Paper I, Theorem (The Hamiltonian realises the isotopy): the one-form $\mathfrak{a}_{i,t}$ is closed because $F_{i,t}$ is symplectic, $\mathfrak{h}_{i,t}$ is its radial primitive, and on $F_{i,t}(\mathbb{B}^\circ(r_{1,i}))$ the field X_H equals $\dot{F}_{i,t}$.

Proposition 2.17 (Hamiltonian of an embedding datum; Paper I)**.** *Let $n \ge 1$, let $(c_i', \bar{c}_i, f_i)$ be an embedding datum for (n, k, C, c), and let $s_1, \dots, s_k$ be centres such that the closed balls $B(s_i, c_i)$ are pairwise disjoint. Then the Hamiltonian $H_{\mathfrak{E}}$ of Definition 2.15 satisfies* $\mathrm{Pack}(n, k, C, c, s)$.

2.5. **Explicitness.**

Definition 2.18 (Levels of explicitness)**.** (L1) An *explicit definitional construction* is a definition built from rational constants and the input by arithmetic and elementary functions, integrals over compact domains, limits of series with an explicit rate of convergence, maxima and minima of continuous functions over explicit compact sets, unique solutions of explicit ordinary differential equations, inverses of explicit injective maps, unique roots of explicit equations, case distinctions on propositions about the input and about the arguments of the function being defined, and least elements of searches over a fixed enumeration of finitely described objects (integers, rationals, and tuples of these and of rational polynomials) whose validity is decided by finitely many explicit inequalities; definite descriptions are used only after unique existence has been proved (Paper I, Definition (Definite description)).

(L2) The *finite form* is $H_{\mathrm{fin}} = \beta_B \cdot P$ with $B \in \mathbb{N}$ and $P \in \mathbb{Q}[t, z]$, where $\beta_B(z) = \prod_{l=1}^{2n} \beta(\zeta_l / B)$ for the real coordinates $\zeta_1, \dots, \zeta_{2n}$ of z, and $\beta(x) = 1 - S\big((x^2 - \frac{1}{4})/\frac{5}{16}\big)$; thus $\beta\colon \mathbb{R} \to [0,1]$ is smooth, equal to 1 on $[-\frac{1}{2}, \frac{1}{2}]$ and to 0 outside $(-\frac{3}{4}, \frac{3}{4})$.

(L3) *Computability* means that one oracle algorithm, given names of the input (a rational input is named by its constant sequence), halts on every admissible input and outputs some (B, P) as in (L2) for which H_{fin} satisfies the packing specification. The output may depend on the names and not only on the numbers they name; this is necessary, as explained below.

The algorithm of Section 5 accepts the first rational polynomial whose Euler trajectories from a finite grid, with an error bound valid on $[0,1]$, stay where the cutoff equals one and end inside the target; every accepted output is correct by a soundness theorem. Termination needs only some Hamiltonian with a positive margin; the existence theorems would also supply one, non-constructively. We use $H[n, k, C, c, s]$.

The dependence on names in L3 is necessary for finite output. A computable single-valued map to the discrete set of polynomial codes is continuous, hence constant on $\varepsilon \in (0,1)$ for fixed k. But the images of $k \ge 2$ disjoint closed balls under any fixed Hamiltonian diffeomorphism form a disconnected compact set, so their density in their smallest enclosing ball is strictly below one, and that fixed output fails the requirement for sufficiently small ε.

3. The models

Theorem 2.6 reduces the packing problem to three inputs: a toric model with handle data, a comparison datum for its profile, and a transfer to the target. This section states the packing

theorem for each model of Table 1: the cubic model of Paper I with general capacities, and the three higher-dimensional models. It then fixes the case distinction that selects them and the embedding datum it produces. The proofs are in §S5 (dimension four), §S9, §S10 (dimension $2n \geq 6$, capacities below one half) and §S11–S14 (one large ball). Throughout, the target is $B^\circ(1)$ and $\rho_i = c_i/C$.

3.1. **Dimension four, general capacities.** Paper I proves the packing in the cubic model for *valid cubic data* (Paper I, Definition (Cubic data)): auxiliary capacities $a_i > \rho_i$, a height $h < \frac{1}{3}$, a number $0 < g \leq 3(1-3h)$, and a comparison function K with a planar Hamiltonian isotopy, the identity near $\pi|z|^2 = h$, satisfying the comparison inequality for the profile $3(1-3p)-\sum_i (a_i-p)_+$ with gap $\frac{31}{32}g$ (Paper I, Theorem (Packing in dimension four)). It constructs such data with a linear K for $k \geq 10$ equal capacities with $\frac{3}{4} < \rho\sqrt{k} < 1$. For $k = 9$ that construction would need $a = h$, and for some unequal inputs no linear K exists (Remark S2.8). The scalar comparison datum covers all inputs: the scalar iteration of §S4.1, an analogue of Proposition 2.9 over real data, with a fixed mollifier, and a planar isotopy built from quadratic wells, an Alexander path and the planar machine of Paper I (Propositions S4.2, S4.7 and S4.8).

Theorem 3.1 (Packing in dimension four)**.** *Let $k \geq 1$ and $0 < \rho_1, \dots, \rho_k < \frac{1}{3}$ with $\sum_i \rho_i^2 < 1$. Then there are explicit numbers $\rho_i < \rho_i^+ < \rho_i^{\mathrm{dom}}$ and smooth maps $f_i \colon B^\circ(\rho_i^{\mathrm{dom}}) \to \mathbb{C}^2$ with symplectic derivative, injective on $B(\rho_i^+)$, such that the sets $f_i(B(\rho_i^+))$ are pairwise disjoint and contained in $B(1-\mu_{\mathrm{ball}})$ for some $\mu_{\mathrm{ball}} \in (0,1)$.*

This is Theorem S5.2. The auxiliary capacities are $a_i = \rho_i + \delta$ with $\delta = \min\big(\frac{1}{8}(\frac{1}{3} - \max_i \rho_i), \frac{D}{8(\sum_i \rho_i + k)}, \frac{1}{8}\big)$, $D = 1 - \sum_i \rho_i^2$, and the height $h < \frac{1}{3}$ is a closed formula in the a_i (Lemma S5.1); the comparison gap is $g = \frac{3}{4}d$ with $d = \frac{1}{2}\min(\bar{P}, 3(1-3h))$, $\bar{P}$ the mean of the profile.

3.2. **Dimension $2n \geq 6$, capacities below one half.** Let $n \geq 3$ and $m = n-1$. The base is a compact piece of the diagonal affine curve

$$\Sigma = \{(t,y) \in \mathbb{C}^{m+1} : \ y_j^2 = t^2 - j, \ 1 \leq j \leq m\}, \qquad \Sigma_\ell = \Sigma \cap \{|t| \leq R_\ell\},$$

with $\lambda = d^c \log(1+|z|^2)$, $\sigma = d\lambda$ and $R_\ell = 2 + \sum_j j + \ell$. The model is $\Sigma_\ell \times \mathbb{C}^m$ with $\Omega_0 = \omega_V + d[(1-2|p|)\lambda]$, $|p| = \sum_j \pi|U_j|^2$, so a ball of capacity a needs $a < \frac{1}{2}$. The base area A_ℓ satisfies $0 \leq 2^m - A_\ell \leq 2^m(m+2)/(\ell+2)$, and the limiting profile $2^m(1-2|p|)$ has integral $1/n!$ over $\{|p| \leq \frac{1}{2}\}$, the volume of $B^\circ(1)$ (Lemmas S9.2 and S9.5). The truncation loss is kept in the comparison profile. The transfer is one map into $B^\circ(1)$: a radial injection, a horizontal transport, an inverse Moser return and an affine ball chart (Theorem S10.12).

Theorem 3.2 (Packing by small balls in dimension $2n \geq 6$)**.** *Let $n \geq 3$ and $0 < \rho_1, \dots, \rho_k < \frac{1}{2}$ with $\sum_i \rho_i^n < 1$, and let $a_1, \dots, a_k$ be real numbers with $\rho_i < a_i < \frac{1}{2}$ and $\sum_i a_i^n < 1$, and put $\rho_i^{\mathrm{dom}} = (\rho_i + 2a_i)/3$. The comparator construction of Theorem 2.11 defines, from these data, smooth symplectic embeddings f_i of open neighbourhoods of the closed balls $B(\rho_i^{\mathrm{dom}}) \subset \mathbb{C}^n$ into $B^\circ(1)$ and a rational $\mu_{\mathrm{ball}} > 0$, such that the sets $f_i(B(\rho_i^{\mathrm{dom}}))$ are pairwise disjoint and contained in $B(1-\mu_{\mathrm{ball}})$.*

This is Theorem S9.7. Construction 3.4 fixes the a_i by explicit slack increments; the further parameters of the model are those of Lemma S9.6.

3.3. **One large ball.** The quadric model accepts only capacities below one half, and the two-ball inequality allows at most one capacity $A \geq \frac{1}{2}$; every other capacity is then below $1-A \leq \frac{1}{2}$. Normalise so that the input is

$$A \geq \tfrac{1}{2}, \qquad A^n + \sum_{i=1}^{k} \rho_i^n < 1, \qquad A + \rho_i < 1 \quad (1 \leq i \leq k). \tag{6}$$

Valid auxiliary data are $(a; r_1, \dots, r_k)$ with $\max(A, \frac{1}{2}) < a < 1$, $\rho_i < r_i < 1-a$ and $a^n + \sum_i r_i^n < 1$; they exist under (6), and rational ones exist as well (Lemma S11.2). The large ball is placed in the reserved ball $B^\circ(a)$ and the others in the shell $S_a = \{a < \pi|y|^2 < 1\}$, of volume $(1-a^n)/n!$; the two-ball condition $A + \rho_i < 1$ is what allows $\rho_i < 1 - a$ with $a > A$.

For $n \geq 4$ the second stage is a product mixed model over a finite bordered piece of the affine quadric curve, with a radial-integral phase lift, and the return to the shell is an explicit slit-shell formula in coupled coordinates (Lemma S12.7). For $n = 3$ that base would be a conic; instead we use the polynomial cubic model of [Ope26], and the large ball is inserted by a polynomial bubble before the final return (§S11.2).

Theorem 3.3 (One large ball)**.** *Let $n \geq 3$, let $A, \rho_1, \dots, \rho_k > 0$, and let $(a; r)$ be valid auxiliary data for $(A; \rho)$. For $n \geq 4$ put $\rho_0^{\mathrm{dom}} = (A+a)/2$ and $\rho_i^{\mathrm{dom}} = (3\rho_i + 5r_i)/8$ $(i \geq 1)$. For $n = 3$ assume moreover $a \in \mathbb{Q}$, let $A < r_0 < a$ and $\rho_i < r_i' < r_i$, and put $\rho_0^{\mathrm{dom}} = r_0$, $\rho_i^{\mathrm{dom}} = r_i'$. Then there are smooth symplectic embeddings f_i of open neighbourhoods of $B(\rho_i^{\mathrm{dom}})$ $(0 \leq i \leq k)$ into $B^\circ(1)$ and a number $\mu_{\mathrm{ball}} > 0$, defined explicitly from these data, such that the sets $f_i(B(\rho_i^{\mathrm{dom}}))$ are pairwise disjoint and lie in $B(1-\mu_{\mathrm{ball}})$. For $n \geq 4$, f_0 is the inclusion, $f_0(B(\rho_0^{\mathrm{dom}})) \subset B^\circ(a)$ and $f_i(B(\rho_i^{\mathrm{dom}})) \subset S_a$ for $i \geq 1$.*

For $n \geq 4$ this is Theorem S12.9; for $n = 3$ it is Theorem S14.7. Construction 3.4 fixes $(a; r)$, and for $n = 3$ also r_i' and r_0, by explicit formulas in ρ.

3.4. **The case distinction.** The following construction is the case distinction of $H[n, k, C, c, s]$; its linear branch is that of Paper I, Construction (The normalised embeddings). For $n = 2$ it uses the cubic model, and for $n \geq 3$ it selects one of the three higher-dimensional models.

Construction 3.4 (The normalised embeddings)**.** Let $(n, k, 1, \rho)$ be admissible. The *normalised embeddings* $f_i^\natural$, enlarged capacities ρ_i^+ and domain capacities ρ_i^{dom} are defined as follows.

(i) If $n = 2$, they are those of Theorem 3.1: the maps of Paper I, Theorem (Packing in dimension four) for the valid cubic data of the linear datum when $k \geq 10$, $\rho_i = \rho$ and $\frac{3}{4} < \rho\sqrt{k} < 1$, with $(\rho_i^+, \rho_i^{\mathrm{dom}}) = (\rho\varrho_3^2, \rho\varrho_4^2)$ of Paper I, Remark (Recorded capacities); and for the valid cubic data of Proposition S4.8 otherwise, with $(\rho_i^+, \rho_i^{\mathrm{dom}}) = (\bar\rho_i, \hat\rho_i)$ of S39.

(ii) If $n \geq 3$ and $\rho_i < \frac{1}{2}$ for all i, put

$$\upsilon_i = \min\left(\frac{1/2 - \rho_i}{2}, \frac{1 - \sum_j \rho_j^n}{2nk}\right), \qquad a_i = \rho_i + \upsilon_i.$$

Use the quadric construction of Theorem S9.7 with these auxiliary capacities. Its enlarged closed capacity and open domain capacity are, respectively, $\rho_i^+ = \rho_i + \upsilon_i/3$ and $\rho_i^{\mathrm{dom}} = \rho_i + 2\upsilon_i/3$.

(iii) If $n \geq 4$ and $\rho_j \geq \frac{1}{2}$, the index j is unique, since $\rho_i + \rho_j < 1$ for $i \neq j$. Write $A = \rho_j$, $r_* = \max(\{0\} \cup \{\rho_i : i \neq j\})$, $V = 1 - \sum_i \rho_i^n$, and put

$$\upsilon = \min\left(\frac{1 - A - r_*}{4}, 1, \frac{V}{2nk2^n}\right), \qquad a = A + \upsilon, \qquad r_i = \rho_i + \upsilon \quad (i \neq j).$$

Use the construction of Theorem S12.9. For the distinguished ball set $(\rho_j^+, \rho_j^{\mathrm{dom}}) = (A + \upsilon/2, A + \upsilon)$; its map is the inclusion, defined and injective on all of $\mathbb{C}^n$, and Theorem 3.3 places $B((A + a)/2) = B(\rho_j^+)$ in $B^\circ(a)$. For $i \neq j$, set $(\rho_i^+, \rho_i^{\mathrm{dom}}) = ((11\rho_i + 5r_i)/16, (3\rho_i + 5r_i)/8)$.

(iv) If $n = 3$ and $\rho_j \geq \frac{1}{2}$, put $A = \rho_j$ and

$$\gamma_0 = \min\left(1 - \sum_i \rho_i^3,\, 1 - A,\, \min_{i\neq j}(1 - A - \rho_i)\right), \qquad \eta = \frac{\gamma_0}{100(k+1)}.$$

The minimum over an empty set is omitted. Set

$$M = \lceil 1/\eta \rceil + 1, \qquad a = \frac{\lfloor (A + 4\eta)M \rfloor + 1}{M}.$$

Use the polynomial construction of Theorem S14.7, with distinguished capacity $r_0 = A + 2\eta$. The auxiliary small family retains all k indices: for $i \neq j$ its outer and inner capacities are $r_i = \rho_i + 3\eta$ and $r_i' = \rho_i + 2\eta$, while the j-th auxiliary ball has input capacity $\eta/4$ and capacities $r_j = \eta$, $r_j' = \eta/2$, so that the small family is nonempty and indexed like the input. The comparison, the weights and the transfer are those of this auxiliary family of k balls, including the j-th ball in every parameter and compact union. The final j-th map is the bubble map of the distinguished ball; the map of the auxiliary j-th ball is not used. Set $(\rho_i^+, \rho_i^{\mathrm{dom}}) = (\rho_i + \eta, \rho_i + 2\eta)$ for every final ball.

For the toric maps in (ii)–(iv), use the finite-atlas normalization and three inverse Moser endpoints of Construction S8.4, applied to the closed source balls $B(\rho_i^{\mathrm{dom}})$. In particular the cap smoothing and layer strip use its internal capacity $(3a_i + \rho_i^{\mathrm{dom}})/4$, with a_i replaced by r_i in cases (iii) and (iv); the distinguished ball, placed by inclusion or by the bubble, is exempt.

Verification of the hypotheses. All increments υ_i, υ, η are positive: admissibility gives $\sum_i \rho_i^n < 1$, $\rho_i < \frac{1}{2}$ in case (ii), and $A + \rho_i < 1$ for $i \neq j$ together with $A < 1$ in cases (iii) and (iv). We use $(x + u)^n - x^n = n\xi^{n-1}u$ for some $\xi \in [x, x + u]$.

Case (ii). $a_i \leq \rho_i + (\frac{1}{2} - \rho_i)/2 < \frac{1}{2}$, and, since $0 \leq \rho_i < a_i < 1$, $\sum_i a_i^n \leq \sum_i \rho_i^n + n\sum_i \upsilon_i \leq \sum_i \rho_i^n + (1 - \sum_j \rho_j^n)/2 < 1$; these are the hypotheses of Theorem S9.7, and $\rho_i^+ = (2\rho_i + a_i)/3$, $\rho_i^{\mathrm{dom}} = (\rho_i + 2a_i)/3$ are the capacities named there.

Case (iii). $(a; r)$ is valid auxiliary data (Lemma S11.2): $a = A + \upsilon > A \geq \frac{1}{2}$; $a < 1$ because $\upsilon < 1 - A$; $r_i + a = \rho_i + A + 2\upsilon \leq r_* + A + \frac{1}{2}(1 - A - r_*) < 1$, i.e. $r_i < 1 - a$; and with $0 \leq x, \upsilon \leq 1$, $(x + \upsilon)^n \leq x^n + n2^{n-1}\upsilon$, so $a^n + \sum_{i\neq j} r_i^n \leq \sum_i \rho_i^n + kn2^n\upsilon \leq \sum_i \rho_i^n + V/2 < 1$. Theorem S12.9 gives $\rho_i^{\mathrm{dom}} = (3\rho_i + 5r_i)/8$ and $\rho_i < (11\rho_i + 5r_i)/16 < \rho_i^{\mathrm{dom}} < r_i$; for the distinguished ball, the inclusion is injective on all of $\mathbb{C}^n$ and $B(A + \upsilon/2) \subset B^\circ(a)$, which is disjoint from the shell S_a containing the other images.

Case (iv). First $1/M < \eta$, so $A + 4\eta < a \leq A + 4\eta + 1/M < A + 5\eta$, and $a \in \mathbb{Q}$. With $\gamma_0 \leq 1 - A$, $\gamma_0 \leq 1 - A - \rho_i$ and $\eta \leq \gamma_0/200$: $a < 1$; $r_0 = A + 2\eta \in (A, a)$; $\rho_i < r_i' < r_i$ and $r_i + a < \rho_i + A + 8\eta < 1$ for $i \neq j$; and for the auxiliary ball $\eta/4 < \eta/2 < \eta$ and $\eta + a < A + 6\eta < 1$. By the mean value bound with $n = 3$ and $x, u \leq 1$, $a^3 \leq A^3 + 60\eta$, $r_i^3 \leq \rho_i^3 + 36\eta$ and $\eta^3 \leq \eta$, so $a^3 + \sum_{i\neq j} r_i^3 + \eta^3 \leq \sum_i \rho_i^3 + (36k + 25)\eta < \sum_i \rho_i^3 + \gamma_0 \leq 1$. Thus $(a; r)$, with the auxiliary j-th entry, is valid auxiliary data with a rational, and Theorem S14.7 applies with r_0 and the r_i'. Its domains are $B(r_0) = B(\rho_j^{\mathrm{dom}})$ and $B(r_i') = B(\rho_i^{\mathrm{dom}})$.

The data may have real coefficients, which is allowed at level L1. The polyhedral construction applies over real coefficients by the observation following Proposition 2.9; the finite-output algorithm certifies its polynomial directly.

Proposition 3.5 (The embeddings)**.** *Let (n,k,C,c) be admissible, $\rho = c/C$, and let $f_i^\natural$, ρ_i^+ and ρ_i^{dom} be given by Construction 3.4. Put*

$$c_i' = C\rho_i^+, \qquad \bar{c}_i = C\rho_i^{\mathrm{dom}}, \qquad f_i(u) = \sqrt{C}\, f_i^\natural(u/\sqrt{C}).$$

Then $(c_i', \bar{c}_i, f_i)$ is an embedding datum for (n,k,C,c), and $\bigcup_i f_i(B(c_i'))$ is a compact subset of $B^\circ(C)$.

Proof. In each case $\rho_i < \rho_i^+ < \rho_i^{\mathrm{dom}}$, and the model theorem gives exactly the properties required on these balls. In case (i), Theorem 3.1 (Theorem S5.2) states them for the displayed pairs. In cases (ii)–(iv) the model theorems (Theorems S9.7, S12.9 and S14.7) give smooth symplectic embeddings of open neighbourhoods of $B(\rho_i^{\mathrm{dom}})$ with pairwise disjoint images of $B(\rho_i^{\mathrm{dom}})$ in $B(1-\mu_{\mathrm{ball}})$: in case (ii) $\rho_i^{\mathrm{dom}} = (\rho_i + 2a_i)/3$; in case (iii) $\rho_i^{\mathrm{dom}} = (3\rho_i + 5r_i)/8$ for $i \neq j$, while the distinguished ball is placed by the inclusion, which is symplectic and injective on all of $\mathbb{C}^n$ and maps $B(\rho_j^+) = B((A+a)/2)$ into $B^\circ(a)$, disjoint from the shell S_a containing the other images; in case (iv) $\rho_i^{\mathrm{dom}} = r_i'$ for $i \neq j$ and $\rho_j^{\mathrm{dom}} = r_0$, with r_i' and r_0 of Construction 3.4. Since $B(\rho_i^+) \subset B^\circ(\rho_i^{\mathrm{dom}}) \subset B(\rho_i^{\mathrm{dom}})$, this is smoothness and the symplectic identity on $B^\circ(\rho_i^{\mathrm{dom}})$ and injectivity, pairwise disjointness and target containment on $B(\rho_i^+)$. Physical rescaling preserves these properties, because $Df_i(u) = Df_i^\natural(u/\sqrt{C})$. The union is a finite union of continuous images of compact sets. □

Definition 3.6 (The Hamiltonian $H[n,k,C,c,s]$)**.** On admissible input with pairwise disjoint closed source balls, let $\mathfrak{E}$ be the embedding datum of Proposition 3.5, together with the centres s_i, and put $H[n,k,C,c,s] = H_{\mathfrak{E}}$ (Definition 2.15).

In the linear branch of case (i) this is the Hamiltonian of Paper I, Definition (The Hamiltonian $H[2,k,C,c,s]$ of the linear branch).

4. Proof of Theorem 1(i), (ii) and of the corollaries

Let (n,k,C,c,s) satisfy the hypotheses of Theorem 1. The normalised capacities $\rho = c/C$ are admissible, and Construction 3.4 selects one of the cases (i)–(iv). In case (i), $n = 2$, the embeddings are those of Theorem 3.1: the cubic model, handle and transfer of Paper I (Paper I, Theorem (Packing in dimension four)) with the linear datum of Paper I or the scalar comparison datum of Proposition S4.8. In case (ii) the embeddings are those of Theorem 3.2, proved in Theorem S9.7: the comparison datum of Theorem 2.11 for the quadric profile minorant of Lemma S9.6, the packing in the model of Theorem 2.6 on the bordered quadric base (Theorem S6.18), and the transfer of Theorem S10.12, composed by Proposition 2.7. In cases (iii) and (iv) they are those of Theorem 3.3, proved in Theorems S12.9 and S14.7. In each case Proposition 3.5 turns them into an embedding datum, and Definition 3.6 defines $H[n,k,C,c,s]$ from it.

Proof of Theorem 1(i). Let $H = H[n,k,C,c,s]$ with the hypotheses of Theorem 1, and let $\mathfrak{E}$ be the datum of Definition 3.6. By Theorem 2.16 (with Paper I, Lemma (Smoothness and support)), H is C^∞, vanishes for $t \notin (0,1)$, and vanishes outside $\mathbb{R} \times K$ with the explicit compact set $K = \mathbb{B}(R_F)$. By Proposition 2.17, H satisfies $\mathrm{Pack}(n,k,C,c,s)$. By Theorem 2.16, $\phi_H^1(B(s_i,c_i)) = f_i(B(c_i))$; these sets are pairwise disjoint and lie in the compact subset $\bigcup_i f_i(B(c_i'))$ of $B^\circ(C)$ (Proposition 3.5). □

Proof of Theorem 1(ii). Every entry of Table 2 is one of the operations of Definition 2.18; the only search in it is the least of $n+1$ angle indices in the linear path. The same definition applies to all (n,k,C,c,s). □

Ingredient	Operation of Definition 2.18	Where shown
hypotheses; branch	case distinctions on propositions ($n = 2$; some $\rho_j \geq \frac{1}{2}$; for $n = 2$, $k \geq 10$, equal $\rho_i = \rho$ and $\frac{3}{4} < \rho\sqrt{k} < 1$)	Def. 3.6, Cons. 3.4
branch margins	arithmetic, finite minima and maxima, square roots, an integer floor and ceiling	Cons. 3.4
model embeddings $f_i^{\natural}$, ρ_i^{+}	model formulas and compact flow parameters; in dimension six, integer weights and explicit polynomial lifts	Thms. 3.1, 3.2, 3.3; Prop. S11.8
comparison datum ($n \geq 3$)	iterates by finitely many Fourier–Motzkin eliminations with N given by a formula; log-sum-exp smoothing	Prop. 2.9, Thm. 2.11
scalar datum ($n = 2$)	finite scalar iteration; convolution with a fixed mollifier (integral over a compact interval)	§S4.1
planar, twist and Moser isotopies	unique solutions of explicit ODEs; inverses of explicit injective maps; unique roots	Thm. 2.10, Thm. S6.18, Paper I, Theorem (Whole-disc Hamiltonian isotopy with prescribed endpoint on the core)
c_i', $\bar{c}_i$, f_i	arithmetic; composition with a dilation	Prop. 3.5
radii $r_{\ell,i}$	square roots; finite minima; arithmetic	Paper I, Lemma (Radii)
linear path $M_{i,\sigma}$	positive square root (definite description); inverses by the adjugate formula; least of $n+1$ angle indices; sin, cos	Paper I, Lemma (The linear path)
L, δ_c, r_{tot}, τ, Y_i	maxima of continuous functions over compact sets; finite minima; arithmetic	Paper I, Lemma (Parameters)
route z_i	ranks (counts of propositions); arithmetic	Paper I, Lemma (The route)
shape, tail, clocks	integral over $[0,1]$; composition; the flat step	Paper I, Lemma (The shape stage), Paper I, Lemma (The tail stage)
$H_{\mathfrak{E}}$	integral over $[0,1]$; radial cutoff; definite description of the unique preimage	Def. 2.15, Paper I, Lemma (Smoothness and support)

TABLE 2. The construction of $H[n,k,C,c,s]$ by the operations of level L1. The finite output of part (iii) adds Bernstein coefficients (finite sums of rational approximations), exact rational Euler tables and the least passing candidate of a decidable search (Section 5).

Part (iii) of Theorem 1 is Theorem 5.3: its termination argument uses only that $H[n,k,C,c,s]$ satisfies $\mathrm{Pack}(n,k,C,c,s)$ with a positive margin, which part (i) provides.

Proof of Corollary 2. Let $n \geq 3$. Suppose Ψ is a symplectic embedding of a neighbourhood of $\bigsqcup_i B(s_i,c_i)$ into $B^\circ(C)$. The image of the compact set $\bigsqcup_i B(s_i,c_i)$ is compact, hence contained in $B(C')$ for some $C' < C$. Symplectic embeddings preserve ω_0^n, so $\sum_i c_i^n/n! = \mathrm{vol}\,\Psi(\bigsqcup_i B(s_i,c_i)) \leq C'^n/n! < C^n/n!$. For $i \neq j$, Ψ restricts to a symplectic embedding of the union of the interiors of $B(s_i,c_i)$ and $B(s_j,c_j)$ into $B^\circ(C'')$ for any $C'' \in (C',C)$, and Gromov's two-ball theorem [Gro85] gives $c_i + c_j \leq C'' < C$. Conversely, if both inequalities hold, then (n,k,C,c) is admissible and Theorem 1(i) applies to $H = H[n,k,C,c,s]$. The global symplectic diffeomorphism ϕ_H^1 maps each $B(s_i,c_i)$ into $B^\circ(C)$, so its restriction to the open neighbourhood $(\phi_H^1)^{-1}(B^\circ(C))$ of $\bigsqcup_i B(s_i,c_i)$ is a symplectic embedding into $B^\circ(C)$. □

Proof of Corollary 3. All $c_i = c$, and $\sum_i c_i^n = kc^n < C^n$. If $n \geq 3$, then $2^n c^n \leq kc^n < C^n$, so $c_i + c_j = 2c < C$. If $n = 2$, then $9c^2 \leq kc^2 < C^2$, so $3c < C$. Thus (n, k, C, c) is admissible and Theorem 1(i) applies. The images are pairwise disjoint and ϕ_H^1 preserves volume, so they fill the fraction $k \operatorname{vol} B(c)/\operatorname{vol} B(C) = kc^n/C^n$ of $B(C)$, and $kc^n/C^n > 1 - \varepsilon$ by hypothesis. □

Proof of Corollary 4. Let $V = \sum_i c_i^2$ and $C = C_\varepsilon$, so $V/C^2 = 1 - \varepsilon/2$. Then $3c_i \leq V^{1/2} < C$, so $(2, k, C, c)$ is admissible. Theorem 1(i) applies, and as for Corollary 3 the images fill the fraction $1 - \varepsilon/2$ of $B(C)$. □

Proof of Corollary 5. Here $B(s_i, \pi)$ is the closed unit ball about $s_i = (3i - 3, 0, 0, 0)$, and these balls are disjoint. Let $C = \pi R^2$ with $R^4 = k/(1 - \varepsilon/2)$. Then $k\pi^2 < C^2$, and $3\pi < C$ since $R^4 > k \geq 9$, so Theorem 1(i) applies. Finally, $B^\circ(\pi R^2)$ is the open ball of radius R, and $k/R^4 = 1 - \varepsilon/2 > 1 - \varepsilon$. □

5. Computability and the finite form

This section proves Theorem 1(iii). The output, a rational polynomial times a fixed smooth spatial cutoff, is found by searching finite certificates. A candidate is accepted when Euler trajectories from a grid covering the sources stay where the cutoff equals one and end inside the target, with an error bound valid on $[0, 1]$. Accepted candidates are correct (Theorem S17.5). Termination uses the Bernstein compilation of $H[n, k, C, c, s]$ (§5.1); any packing Hamiltonian with a positive margin would serve, and for the Epoch family the witness is Corollary 5 at density $1 - \varepsilon/2$. A name for a real x is a rational stream $q(p)$ with $|q(p) - x| \leq 2^{-p}$; rationals are named by constant streams. Names of $r_i = \sqrt{c_i/\pi}$ and $R = \sqrt{C/\pi}$ are computed in §S15. Here $|\cdot|$ is the Euclidean norm and S is the flat step of Definition SA.1.

5.1. **The finite form.**

Lemma 5.1 (Grönwall)**.** *Let $G, H \colon \mathbb{R} \times \mathbb{R}^{2n} \to \mathbb{R}$ be C^∞, let $\eta > 0$ and $L_X > 0$, and suppose that*

(i) $|X_G(t, z) - X_H(t, z)| \leq \eta$ for all $t \in [0, 1]$ and all $z \in \mathbb{R}^{2n}$;
(ii) $X_H(t, \cdot)$ is L_X-Lipschitz on all of $\mathbb{R}^{2n}$ for every $t \in [0, 1]$.

If $\gamma_G, \gamma_H \colon [0, 1] \to \mathbb{R}^{2n}$ are integral curves of X_G and X_H with $\gamma_G(0) = \gamma_H(0)$, then $|\gamma_G(1) - \gamma_H(1)| \leq \eta(e^{L_X} - 1)/L_X$. If moreover $\eta = b_R e^{-L_X}/2$ with $b_R > 0$, then $\eta(e^{L_X} - 1)/L_X < b_R$.

Proof. $u = \gamma_G - \gamma_H$ satisfies $u(0) = 0$ and $|\dot{u}| \leq \eta + L_X|u|$, so Grönwall gives the bound. With $\eta = b_R e^{-L_X}/2$ it equals $\frac{b_R}{2}(1 - e^{-L_X})/L_X < b_R/2$ (cf. Lemma S16.14). □

For smooth compactly supported H, (ii) holds for every $L_X > 0$ with $L_X \geq \max_{[0,1]\times\mathbb{R}^{2n}} \|D_z^2 H\|$, as $X_H = J_0 \nabla_z H$ with J_0 orthogonal.

Let $n \geq 2$, $k \geq 1$, $C > 0$, $c \in (0, \infty)^k$, $s \in (\mathbb{R}^{2n})^k$, and let H satisfy $\mathrm{Pack}(n, k, C, c, s)$ (Definition 1.1): H is C^∞, vanishes outside $\mathbb{R} \times K_H$ with K_H compact, and $\phi_H^1(B(s_i, c_i)) \subset B^\circ(C)$. Write $\zeta_1, \ldots, \zeta_{2n}$ for the coordinates of z. A *compilation datum* for H consists of:

(C1) $C_0 \in (0, C)$ with $\phi_H^1(B(s_i, c_i)) \subset B(C_0)$ for all i; put $R_* = \sqrt{C_0/\pi} < R_C = \sqrt{C/\pi}$.
(C2) A rational $a_{\mathrm{box}} > 0$ with $H(t, z) = 0$ whenever $\max_l |\zeta_l| \geq a_{\mathrm{box}}$, and an integer $B > \max(2n, 2a_{\mathrm{box}})$; the box cutoff is

$$\beta_B(z) = \prod_{l=1}^{2n} \beta_{B,1}(\zeta_l), \qquad \beta_{B,1}(x) = 1 - S\Big(\frac{x^2 - (B/2)^2}{(3B/4)^2 - (B/2)^2}\Big),$$

so $0 \leq \beta_B \leq 1$, $\beta_B = 1$ on $[-B/2, B/2]^{2n}$, $\beta_B = 0$ outside $(-3B/4, 3B/4)^{2n}$, and $\beta_B H = H$.

(C3) Reals $L_X \geq 1 + \sup_{[0,1]\times\mathbb{R}^{2n}} \|D_z^2 H\|$ and $C_\beta \geq 1 + \sup_{\mathbb{R}^{2n}} |\nabla\beta_B|$.
(C4) A rational b_R with $0 < b_R < (R_C - R_*)/4$; put $\eta = b_R e^{-L_X}/2$.
(C5) A real $M_0 \geq 1$ bounding all first and second partials on $[0,1]^{2n+1}$ of $\tilde{H}(\xi) = H(\xi_0, 2B\xi_1 - B, \ldots, 2B\xi_{2n} - B)$.
(C6) A rational δ with $0 < \delta < \min\bigl(\frac{1}{2}, \eta/(16(1+C_\beta))\bigr)$ and an integer $N_{\mathrm{B}} > \max\bigl(1, ((2n+1)M_0/\delta)^2, 4M_0/\delta\bigr)$.
(C7) For every $m \in \{0,\ldots,N_{\mathrm{B}}\}^{2n+1}$ a rational $\tilde{H}_m$ with $|\tilde{H}_m - \tilde{H}(m/N_{\mathrm{B}})| < \delta/(2N_{\mathrm{B}}+1)$.

With $b_{N,j}(x) = \binom{N}{j} x^j (1-x)^{N-j}$ put

$$P(t,z) = \sum_{m\in\{0,\ldots,N_{\mathrm{B}}\}^{2n+1}} \tilde{H}_m \, b_{N_{\mathrm{B}},m_0}(t) \prod_{l=1}^{2n} b_{N_{\mathrm{B}},m_l}\Bigl(\frac{\zeta_l + B}{2B}\Bigr), \qquad H_{\mathrm{fin}} = \beta_B \cdot P,$$

so $P \in \mathbb{Q}[t,z]$ has degree at most N_{B} in each variable. Compilation data exist: $\bigcup_i \phi_H^1(B(s_i,c_i))$ is a compact subset of $B^\circ(C)$, H has compact support, (C3), (C5) are maxima of continuous functions on compacta, and the other items are strict bounds on single numbers.

Theorem 5.2 (Finite output)**.** *Let H satisfy* $\mathrm{Pack}(n,k,C,c,s)$*. For every compilation datum (C1)–(C7) for H, the function $H_{\mathrm{fin}} = \beta_B \cdot P$ satisfies* $\mathrm{Pack}(n,k,C,c,s)$*. More precisely: H_{fin} is C^∞ and $H_{\mathrm{fin}}(t,z) = 0$ for all t whenever $z \notin (-3B/4, 3B/4)^{2n}$; $|X_{H_{\mathrm{fin}}} - X_H| < \eta$ on $[0,1]\times\mathbb{R}^{2n}$; and for every $z \in \bigcup_i B(s_i,c_i)$,*

$$|\phi^1_{H_{\mathrm{fin}}}(z) - \phi^1_H(z)| < b_R, \qquad |\phi^1_{H_{\mathrm{fin}}}(z)| < R_* + b_R < R_C,$$

so that $\phi^1_{H_{\mathrm{fin}}}$ maps every $B(s_i,c_i)$ into $B^\circ(C)$. If the source balls are pairwise disjoint, their images are pairwise disjoint.

Proof (complete version: Theorem S16.15). The tensor Bernstein estimates (Proposition S16.7) and the rounding (C7), harmless since Bernstein weights are nonnegative with sum 1, give $|P - H| < \delta$ and $|\nabla_z P - \nabla_z H| < \delta$ on $[0,1]\times[-B,B]^{2n}$ (Lemma S16.9). As $\nabla_z H_{\mathrm{fin}} - \nabla_z H = \beta_B \nabla_z (P-H) + (P-H)\nabla\beta_B$ there, its norm is $< \delta(1+C_\beta) < \eta/16$ by (C6); outside $(-B,B)^{2n}$ both vanish nearby. Hence $|X_{H_{\mathrm{fin}}} - X_H| < \eta$, X_H is L_X-Lipschitz by (C3), the flows are global, and Lemma 5.1 with (C1) gives the estimates. Support holds as β_B is independent of t; disjointness follows from injectivity of $\phi^1_{H_{\mathrm{fin}}}$. □

5.2. **The algorithm.** Put $d = 2n$. A candidate consists of integers $B, Q, N \geq 1$, precisions $p, m \geq 0$, and a rational arithmetic expression e for a polynomial $P_e(t,z)$. The integer $\Theta = h_B(e)$ (Definition S17.1) bounds, in maximum norm, the speed and the temporal and spatial Lipschitz constants of

$$F_e(t,z) = J_0 \nabla_z P_e(\mathrm{clip}_{[0,1]}\, t, \mathrm{clip}_{[-B/2,B/2]^d}\, z),$$

clip_X being nearest-point projection to the box X. Read the input names to error 2^{-p}; form rational outer balls $B(s_i^p, r_{i,p}^+)$ containing the closed sources and a rational $R_p^- \leq \sqrt{C/\pi}$ (Equation S83); test $|s_{i,l}^p| + r_{i,p}^+ < Q$ for all i,l. On the grid of mesh $\rho = 2Q2^{-m}$ in $[-Q,Q]^d$, discard (node a, source i) if $|a - s_i^p|^2 > (r_{i,p}^+ + d\rho)^2$; otherwise compute the exact Euler table

$$x_0 = a, \quad x_{j+1} = x_j + N^{-1} F_e(j/N, x_j), \quad E = \left(\rho + \frac{\Theta + \Theta^2}{N}\right) 4^\Theta + \frac{\Theta}{N},$$

and require $|x_{j,l}| + E < B/2$ for all $j \leq N$ and l, $R_p^- - dE > 0$ and $|x_N|^2 < (R_p^- - dE)^2$. These are finite strict rational tests; the algorithm returns B and $P_e \in \mathbb{Q}[t,z]$ for the first candidate passing all of them. The real inputs are oracles and n, k ordinary input (§S18). The returned

polynomial need not be the Bernstein approximation of $H[n,k,C,c,s]$: Bernstein compilation only shows that some candidate is accepted, and every accepted candidate is correct.

Theorem 5.3 (Termination and correctness)**.** *There is one oracle machine with the following property. Let $n \geq 2$, $k \geq 1$, let (n,k,C,c) be admissible and let the closed balls $B(s_i,c_i) \subset \mathbb{R}^{2n}$ be pairwise disjoint (these are the hypotheses of Theorem 1). Supply as oracles names of $C, c_1, \ldots, c_k$ and of the $2nk$ coordinates of $s_1, \ldots, s_k$. On input (n,k) the machine halts and outputs an integer B and a polynomial $P \in \mathbb{Q}[t, z_1, \ldots, z_{2n}]$ such that $\beta_B P$ satisfies* $\mathrm{Pack}(n,k,C,c,s)$*. The output may depend on the names, not only on the numbers they name. If the names are computable, for instance the constant names of rational inputs, replacing the oracles by programs for them gives an ordinary algorithm.*

There is also one oracle machine which, given a name of $\varepsilon \in (0,1)$ as oracle and an integer $k \geq 9$ as input, computes names of the source radius 1 *and of the target radius $R = (k/(1-\varepsilon/2))^{1/4}$ for the data*

$$n = 2, \quad C = \pi R^2, \quad c_i = \pi, \quad s_i = (3i-3, 0, 0, 0),$$

runs the same certificate search on them, and halts with (B,P) such that $\beta_B P$ satisfies $\mathrm{Pack}(2,k,C,c,s)$*; here $k/R^4 = 1 - \varepsilon/2 > 1 - \varepsilon$.*

Proof. Correctness. Each source point is within ρ (maximum norm) of a retained node and the Euler error is at most E, so the plateau tests keep the F_e-trajectory where $F_e = X_{\beta_B P_e}$ and the terminal test ends it in the open target (Theorem S17.5).

Termination. $H[n,k,C,c,s]$ maps the sources into $B^\circ(C_0)$, $C_0 < C$. Theorem 5.2 with target $(C_0 + C)/2$ and B large gives a rational polynomial whose trajectories from a neighbourhood of the sources stay a positive distance inside $\{\beta_B = 1\}$ and end a positive distance inside $B^\circ(C)$; as retained nodes lie within $(2d+3)2^{-p} + d\rho$ of a source, all tests pass for p, m, N large (Theorem S17.7). Each candidate uses finitely many queries and operations, and n, k are inputs of one machine (§S18).

Epoch case. The input is admissible with disjoint sources (Corollary 5); radius names come from ε by total programs: the reciprocal of $1 - \varepsilon/2 > \frac{1}{2}$ and two square roots (Lemma S15.6). □

Remark 5.4 (Uniformity and size). The exact construction distinguishes the four cases of Construction 3.4, but the finite verifier has the same form in every case: it tests rational trajectories of the proposed output, and needs no equality test between named capacities. The factor 4^Θ in the Euler bound can require very large grids; the theorem concerns computability, not running time.

Remark 5.5 (The Epoch submission format). Epoch [Epo26] asks for a Python function returning a PyTorch implementation of the Hamiltonian, which a numerical verifier integrates from sampled boundary points of the source balls within fixed time limits. We read its variables as $(Q,P) = (x,y)$, with $X_H = (\partial_P H, -\partial_Q H)$, and its target as the time-one map. The construction and the algorithm are designed for proof and computability, not for integration within the verifier's limits of ten minutes per solution and ten seconds per evaluation. The repository includes two accompanying programs: one for the Epoch Hamiltonian interface, based on the elementary constructions of the third paper of the release over a covered range of (k,ε), and one evaluating closed-form C^4 packing embeddings directly for all $k \geq 10$ and $\varepsilon \in (0,1)$ (proved in the fourth paper of the release).

6. Formal verification

The statements of this paper are verified in the Lean 4 development shared with Paper I [Son26b, dMU21], with the toolchain `v4.34.1`, Mathlib [The20] at the tag of the same name,

and the unmodified Lean library accompanying [Ope26]. That library contributes lemmas and a few generic differential-geometric operators (exterior derivative, chart differential); the definitions of the Hamiltonian and of the algorithms belong to the development, and none of the library's packing constructions or existence results is used. In the directory `verification/`, `Problem.lean` defines the problem using Mathlib alone, `Algorithm.lean` exposes the top level of the construction, `Statement.lean` states its correctness, and `Computability.lean` and `EpochComputability.lean` state correctness and termination of the finite-output algorithms. Theorems in `verification/` have no namespace; other names are relative to `SymplecticPacking`. The statement files, the trusted base and the metaprograms are described in Paper I, § "Formal verification".

Statement	Lean
Definitions 1.1, 1.2	`Problem.PackingSpec`, `Problem.Admissible`
Theorem 1(i)	`main_theorem`, `main_time_support`, `main_support_radius`; for the flow, `flow_is_integral_curve`, `flow_symplectic_diffeomorphism`, `flow_images_disjoint`
Theorem 1(ii)	`#check_explicit General.packingHamiltonian` (a metaprogram)
Theorem 1(iii)	`original_name_algorithm`, with `#check_constructive` on the machine; finite form: `General.Compile.main_finite`
Corollary 2	`neighbourhood_criterion`, `flow_neighbourhood_embedding`
Corollary 3	`equal_balls`
Corollary 4	`unequal_cone`
Corollary 5	`epoch_packing`
Theorem 5.3	`original_name_algorithm`, `epoch_name_algorithm`, `epoch_name_algorithm_packing`
embeddings, by case	`Branch2.embeddingsEq_spec`, `Branch2.embeddingsGen_spec` ($n = 2$, linear and scalar datum), `BranchSmall.embeddings_spec` (all $c_i < \frac{C}{2}$), `BranchLargeHigh.embeddings_spec` ($n \geq 4$, one $c_j \geq \frac{C}{2}$), `BranchLarge3.embeddings_spec` ($n = 3$, one $c_j \geq \frac{C}{2}$)
transfers	`BranchSmall.vectorTransfer_spec`, `BranchLargeHigh.shell_pullback`, `BranchLarge3.cubicTransfer_spec`
comparison and handle	`HighDim.CmpInput.comparatorDatum`, `HighDim.liftGen_spec`, `HighDim.RawModelAgreement.native_packing_spec`, `HighDim.assembly_spec`

TABLE 3. Formal counterparts of the statements of this paper. Every case of Construction 3.4 is covered by `main_theorem` and the algorithm theorems.

The statement. Phase space is `Problem.E` n = `EuclideanSpace` $\mathbb{R}$ (`Fin` $(2n)$); `omega0` and `hamVF` give ω_0 and $X_H = (\partial_y H, -\partial_x H)$, so $\iota_{X_H}\omega_0 = dH$. `capBall` s $c = \{z : \pi\|z - s\|^2 \leq c\}$ and `capBallOpen` is its strict version. `Admissible`, `BallsDisjoint` and `PackingSpec` are Definitions 1.2 and 1.1 and the disjointness hypothesis, field by field; `PackingSpec` quantifies over every integral curve on $[0, 1]$, so it does not depend on a chosen solution. `Algorithm.lean` shows by `rfl` that `General.packingHamiltonian` is `HamEnd.hamiltonianOfEmbeddings` applied to `General.selectEmbeddings`, the case distinction of Construction 3.4. The Lean statements agree with the text, with these differences of form: `equal_balls` does not assume $\varepsilon > 0$; the algorithm theorems assert `Nat.RecursiveIn` relative to the oracles given by the input names, and that one program serves every input is certified by `#check_constructive`; `flow_neighbourhood_embedding` uses the neighbourhood $(\phi_H^1)^{-1}(B^\circ(C))$ and holds for every $n \geq 2$.

What is verified. The kernel checks the theorems of `Statement.lean` and `Computability.lean` and all their dependencies in the development, the vendored library and Mathlib. The trusted base is the kernel, the definitions in `Problem.lean` and the axioms `propext`, `Classical.choice` and `Quot.sound`; the build fails if the transitive axiom set of any of these theorems contains anything else, in particular `sorryAx`. Explicitness (level L1) and uniformity (level L3) are checked by the metaprograms `#check_explicit` and `#check_constructive`, which are not themselves verified. The first is described in Paper I, § "Formal verification" and Paper I, § "The explicitness check"; it is also applied to the algorithm terms. The second, applied to the two oracle machines, additionally forbids any classical constant, including classical decidability, so each is one fixed program in its oracle streams and cannot branch on an undecidable property of the named reals. `test/ExplicitTest.lean` contains terms both commands must reject, among them a machine branching on $x < \frac{1}{3}$ for real x.

Independence. The command `#assert_construction_independence` in `verification/Audit/Independence.lean` traverses types, definitions and proof bodies of the packing, flow, corollary and algorithm theorems and fails if they reach the vendored packing predicate or existence theorems, the holomorphic-line and two-ball obstruction proofs, or the curve-analysis modules of the vendored library. In particular the sufficiency direction of Corollary 2 does not use the packing existence theorem of [Ope26]. The converse is checked separately: it may use the classical obstructions, but no sufficiency theorem.

Source tags. The sources of the main text and the supplement attach Lean declaration names to further statements; `verification/PAPER-INDEX.md` indexes them, and the build checks that every tagged name exists and depends only on the three axioms. A tag is a reference to related declarations, without a claim that the Lean statement covers the printed one, except as follows. Definitions 1.1 and 1.2, Theorem 1(i), (iii) and Corollaries 2–5 are verified in full, up to the differences of form above; Theorem 1(ii) is checked by `#check_explicit`. For Theorem 2.10 (Theorem S3.13), Theorem SA.13 and Lemma SA.18 the formal statement is the consequence used by the construction, not every assertion or the whole domain of the printed statement; the restrictions are recorded in Paper I, § "Formal statements that differ in form from the text". Table 4 lists the Lean declarations for the checking route of §S1.

Table 4: Lean declarations for the steps H1–H7 of §S1 and the steps F1, F9–F11 of Paper I (Paper I, § "The four-dimensional chain"). Names are relative to `SymplecticPacking`, except the theorems of `verification/`, which have no namespace.

step	Lean declarations
F1	`General.selectEmbeddings`, `Branch2.EqualCase`
F9	`General.Ham.Isotopy.IsoInput.commonIsotopy`, `…isoF_sympl`, `…isoF_disj`
F10	`HamEnd.hamiltonianOfEmbeddings`, `General.Ham.hamVF_ofIsotopy_image`, `HamEnd.hamiltonian_end`
F11	`main_theorem`, `four_dim_theorem`, `epoch_packing`, `epoch_theorem` in `verification/Statement.lean`
H1	`HighDim.CmpInput.comparatorDatum`, `HighDim.ComparisonDatum`, `HighDim.PlanarGen.Input`
H2	`Branch2.cmpInput`, `Branch2.cubicDataOf`, `Branch2.Hyp.cubicDataOf_valid`, `Branch2.embeddingsGen_spec`
H3	the tags of Theorem S6.18 in `PAPER-INDEX.md`; composition `HighDim.assembly_spec`
H4	`BranchSmall.vectorTransfer_spec`, `BranchSmall.embeddings_spec`
H5	`BranchLargeHigh.shell_pullback`, `BranchLargeHigh.embeddings_spec`

Continued on the next page

Table 4 (continued)

step	Lean declarations
H6	`BranchLarge3.cubicTransfer_spec`, `BranchLarge3.embeddings_spec`; corollaries `main_theorem`, `neighbourhood_criterion`, `equal_balls`
H7	`General.Compile.main_finite`, `original_name_algorithm`, `epoch_name_algorithm`

Authorship and the role of AI

Liang Song is the sole author and is responsible for the manuscript and its release. The mathematics, the Lean formalisation and the text were produced by AI systems, including GPT-6.1 Sol (OpenAI), Claude Opus 5.5 (Anthropic) and other models, working in a research environment designed and directed by Liang Song, who chose the problem, assembled the literature and context given to the systems, designed the verification pipeline and made the editorial decisions. The scope of the formal verification is described in Section 6.

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